SOLUTION DIAGRAM & ARCHITECTURE

Turn retiring experts’ judgement into rules AI can’t override.

Keystone compiles what a senior engineer knows into an **Intelligence Pill** — a versioned, tested, plain-text artefact the organisation owns, with deterministic gates no model can talk past.

`keystone-tencent.vercel.app`
live demo · passcode `keystone`

`github.com/frederickemerson/keystone`
source, pills and tests

THE REAL-WORLD PROBLEM

The judgement that keeps a building safe is in a few heads, and those heads are leaving.

1

Concentrated

Keppel's own brief: critical knowledge sits with a few experts; new staff need a long time to build it.

- One Chief Engineer, 18 years of M&E, holds the chilled-water leak playbook.
- A wrong call means unsafe re-energisation.

2

Never written down

The useful part is tacit: when to break the rule, and why.

- "Always isolate the riser" **vs** "isolate the sub-valve" — both true, under different conditions.
- SOPs record the step, not the judgement.

3

Lost at retirement

Replacing an employee costs 0.5–2× annual salary (Gallup); FM talent growth "is not keeping pace" (JTC, 2025).

- The knowledge leaves with the person.

4

AI can't be trusted to act

A chatbot that answers from a manual will also answer when it shouldn't.

- Asked outside its pill, our published agent replies: *"This is not covered by an approved pill."*

Who this is for

REIT managers and property/facility-management organisations running mixed portfolios — offices, data centres, logistics. We model the real chain: REIT manager → property/facility manager → FM contractor, with 15 named personas.

Pain points sourced from the Keppel brief, Investor Day 2026, Keppel REIT SR24, Gallup and JTC

Keystone

Spoken judgement becomes a governed Intelligence Pill

Solution diagram · Keppel AI HARVEST
Pill in focus: Technical Services, chilled-water leak · fictional data

1 Business problem

Incident-response judgement sits with a few senior engineers, is applied inconsistently and leaves when they retire. A wrong call on a leak means unsafe re-energisation, tenant/SLA exposure and downtime. A chatbot can't be trusted to act.

2 Capture what was never written

Whose: Tan Wei Ming, retiring Chief Engineer, 18 years of M&E.

- Onboarding call with **CDM + ACTA probes** (simulated in the demo): "walk me through the last time…"
- **Contradiction resolution:** "always isolate the riser" vs "isolate the sub-valve" becomes one condition (TS-02)
- **Expert confirms, edits or rejects** every rule on the Decision Canvas

3 Compile into a reusable pill

A plain-text package the organisation owns:

`SKILL.md` `rules.yaml` `evidence.yaml` `provenance.json` `evals/` `CHANGELOG.md`

- 13 rules, each with action class, owner, deadline and expert quote + timestamp
- Owner, steward and approver named per pill
- Runs as a **native CodeBuddy skill** (verified): same TS-07 decision

4 Asset Operations Manager supervises

Incident Room: a deterministic engine applies the pill to evidence.

Recommend · Requires approval · Must escalate · Blocked

- **Verify:** Why drawer shows rule, version, missing evidence and the expert's quote, read aloud
- **Escalate:** TS-07 blocks re-energisation and routes to the duty LEW by name
- **Approve:** named humans only; server-side role-based redaction

5 More than a chatbot

- **Deterministic gates** an LLM cannot talk past
- **Provenance** to the spoken sentence
- **Evals** replay past incidents
- **Governance:** versions, owners, audit, rollback
- Engine takes no model input: an LLM change can't alter a gate

6 Governed learning after deployment

Evals pass + an approver who neither wrote nor shadowed it signs. History is append-only.

Feedback	Review	Diff	Validate	Shadow	Approve	Release
Outcome captured: isolated in 7 min	Expert corrects TS-02 deadline 15 → 10 min	v1.2.0 → v1.3.0 draft, reason recorded	Replay evals on past incidents; regressions block	Steward replays 6 closed incidents on both versions: 6/6 agree	Three people: author ≠ steward ≠ approver; enforced in code	New semver version; one-click rollback

7 Extend across assets and units

- **Asset overrides:** on Harbourfront DC1 the same pill escalates to the Critical Facilities Manager
- **Coordinate:** Tenant Experience pill runs beside Technical Services
- **Registry:** coverage and gaps by asset type and domain

8 Credible implementation path on Tencent Cloud

Running today: Next.js app on Vercel (live AI there: hy4-preview via TokenHub, then OpenAI, else recorded), deterministic engine, CodeBuddy skill (verified)

CodeBuddy / WorkBuddy	Hunyuan models	ADP	iBMS · CMMS	TRTC voice	EdgeOne · SSO
Pill as SKILL.md skill (verified); WorkBuddy Experts proposed	hy4-preview via TokenHub (live, ap-singapore) and CodeBuddy SDK (local)	Proposed: governed skill distribution, Approval Center	Proposed: leak, alarm and work-order feeds (Brick 1.5); mock today	Proposed: live onboarding call with captions	Proposed hosting (Vercel today); enterprise SSO for roles

BUSINESS ARCHITECTURE

One loop, five stages, a named human at every gate.

Capture

Teach a skill

The expert answers one wide question, then only what is still missing.

- Answers are taps, so a **contradiction is arithmetic**, not a model's opinion.
- A frontline note at the moment a gate blocks someone feeds the same loop.

Tan Wei Ming · Rajesh Kumar

Compile

Confirm the rule

The AI only drafts. Every rule shows the sentence it came from, with a timestamp.

- The expert confirms, edits or rejects each one on the Decision Canvas.

Tan Wei Ming

Apply

The gate holds

A deterministic engine applies the pill to the evidence of a live incident.

- TS-07 blocks re-energisation and escalates to the **duty Licensed Electrical Worker**, by name.

Priya Nair

Learn

Correct it safely

Diff → eval replay on past incidents → shadow → approval → one-click rollback.

- **Three different people**: author ≠ steward ≠ approver, enforced in code.

Hafiz · Priya · Goh

Reuse

Across the portfolio

The same pill runs at a colocation data centre through asset overrides.

- Escalation reroutes to the Critical Facilities Manager; the notice becomes a client-SLA notice.

Every asset of that type

The loop closes

Reuse feeds the registry, which shows coverage and gaps by asset type and domain — so the next gap is visible before the next expert retires. Every stage in this row is built and working in the demo; nothing here is a mockup.

THE REUSABLE UNIT

An Intelligence Pill is plain text the organisation owns.

What is in the package

`pills/technical-services/chilled-water-leak/`

`SKILL.md`	Owner, steward, approver, version, access and boundaries — plus the prose an LLM host reads.
`rules.yaml`	**13 rules**: structured triggers, required evidence, action class, forbidden actions, escalation target, deadline, asset overrides.
`evidence.yaml`	The shared vocabulary — one contract both the pill and the app are tested against.
`provenance.json`	Every rule mapped to the expert's quote, timestamp and confirmation.
`evals/`	Six past incidents a new version must still agree with.
`CHANGELOG.md`	Append-only history.

Why plain text matters

- **Portable.** Copied unchanged into `.codebuddy/skills/`, CodeBuddy reached the same TS-07 decision as our engine — blocked, escalate to the duty LEW, quoting the Chief Engineer. **Verified, not claimed.**
- **Reviewable.** A rule change is a diff an engineer can read.
- **Model-independent.** The decision path takes no model input, so swapping the model cannot change a safety outcome.
- **Yours.** It is a folder in your repository, not rows in a vendor's database.

The gate, in one line

Status precedence is fixed code:

`not_applicable` → blocked → escalate → needs approval → allowed

- Evidence counts only when it is confirmed and exactly true.
- An exception may **add** a requirement. It can never relax a gate.
- No AI proposal ever executes. Each is held for a named human.

Two pills, one incident

A second pill — `tenant-experience/incident-comms` — runs beside it during the same leak: it drafts the tenant notice under its own hard rules while Technical Services works the gate.

TECHNICAL ARCHITECTURE

A deterministic core, with the model kept outside it.

Surfaces

Next.js 15 App Router · React 19 · TypeScript · Tailwind v4 · Vercel

- Teach a skill · Expert call · Decision Canvas
- Incident Room · Registry · Approvals · Learning
- Frontline capture from a gate block, job close-out or a QR label on the plant

`lib/engine`

Pure TypeScript. No LLM, no `eval()`, no string expressions.

- Structured conditions: `eq, neq, gt, gte, lt, lte, in, exists`
- Returns actions, an evidence count (never an invented percentage) and a trace
- Counterfactuals: “what would change this decision?”

`lib/governance`

Versions, approvals, audit.

- Semantic versions, JSON diff, append-only rollback
- Three-person separation of duties, enforced — and each refusal audited
- Role and sensitivity-based redaction, applied on the server before data reaches the browser

`lib/llm` · Tencent Cloud

Language work only. Never a decision.

- **Hunyuan hy4-preview via TokenHub** — live on the public demo, dedicated Singapore endpoint (~3.9 s)
- **MiniMax Speech 2.8 HD via TokenHub** — the voice on screen and the film’s narration
- Hunyuan through the CodeBuddy Agent SDK locally; OpenAI then recorded output as fallbacks
- The badge always names the provider that actually answered

Proven, not asserted

1,302 automated tests, including adversarial and property-based tests (fast-check) over the engine and the governance layer. Design mapped to the IMDA Model AI Governance Framework (2020 and the 2026 Agentic AI edition), AI Verify, ISO/IEC 42001 and the NIST AI RMF.

Proposed next on Tencent Cloud: ASR in Singapore, ADP distribution, TRTC voice interview, iBMS/CMMS evidence feeds, EdgeOne hosting, enterprise SSO

HOW PROMPTS DRIVE THE AI GENERATION

The model does language. It never decides a safety action.

1

Elicit

One wide question first — “walk me through the last time…” — because an unprompted account gives up more than ten narrow questions do.

- Probes tagged against a 31-probe library from the Critical Decision Method and ACTA.

2

Extract

A schema-constrained “scribe” prompt maps speech into the `rules.yaml` schema.

- Strict structured output, thinking disabled.
- Output is always a **DRAFT candidate** the expert must confirm.
- If no live model answers, it says so — it will not show a recorded reading of a different conversation.

3

Resolve

Two answers about the same field with different claims disagree.

- In the exercise this is **string comparison, not inference** — the engine simply knows.
- The conflict becomes one distinguishing condition; neither half is dropped.

4

Compose

Tenant notices are drafted under the Tenant Experience pill’s hard rules.

- No cause before the cause is confirmed (TE-03); no completion time before isolation (TE-02); PII masked (TE-05).
- Unsupported prose fails closed. A named human approves. Keystone never sends it.

The invariant

The decision path takes no model input. Change the model, change the wording — you cannot change whether the gate holds.

Rule drafting, contradiction resolution and notice drafting are all demonstrable in the live app

BUSINESS VALUE

What it is worth, and how we would measure it.

0.5–2×

annual salary to replace an employee (Gallup) — and with them, the judgement nobody wrote down.

>US$100k

the cost of more than half of significant data-centre outages; about 1 in 5 cost more than US$1M (Uptime Institute).

S$125k

a month: one 10,000 sq ft Grade A floor at S$12.50 psf (CBRE Q2 2026). A two-week closure is ~S$58k in rent alone.

What Keystone already measures, over the demo data

- **Time to isolate**, read from the incident records.
- **Replay agreement** — "x of y assertions", from the deterministic engine, never a made-up percentage.
- **Eval pass and shadow agreement** per pill; decisions awaiting approval; pills approved; pending corrections.
- An illustrative **downtime-avoided** estimate, with its assumption stated on screen (4 hours per blocked re-energisation).

The pilot would measure

- Time to isolate · escalation latency · gate-block rate · **override rate** · eval agreement · rollback frequency.
- Goal: a measurable cut in time to isolate and in unplanned re-energisation risk — and new engineers making guided decisions with named escalation **from day one**, instead of after months of shadowing.

Anchored to Keppel's own goal

Technical Services aims to "cut downtime and costs" (Investor Day 2026). Keppel DC REIT reports ~12% better effective PUE on asset-enhanced colocation assets (AR2025) — the gains our data-centre overlay is built to protect.

Figures are external and cited; everything measured over demo data is computed from the records, not asserted